

SET 2: CUSTOMER FEEDBACK ANALYSIS

Dataset: Customer Satisfaction

Department	AIML
Course Code / Course Name	DSA0602/ Data Handling and Visualization
Experiment	Data Visualization using R
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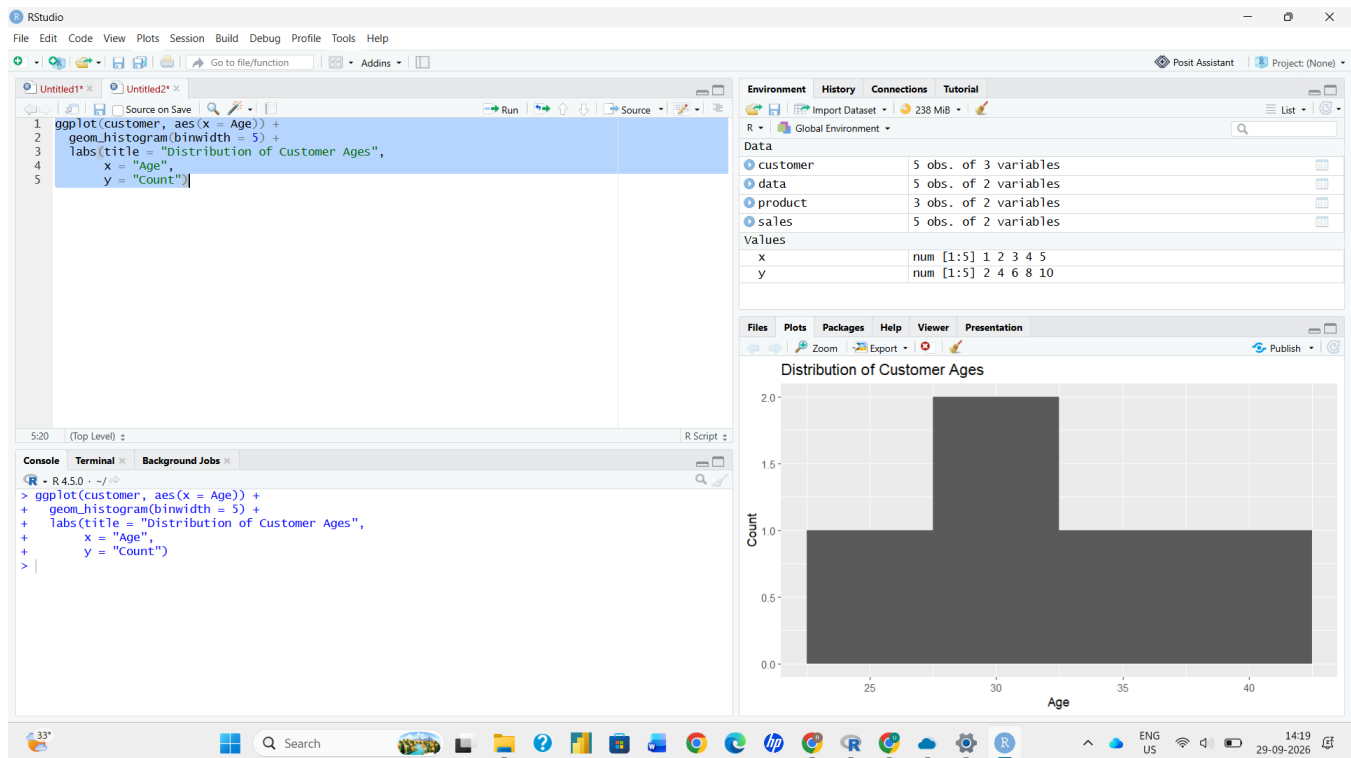
Customer ID	Age	Satisfaction Score
1	25	4
2	30	5
3	35	3
4	28	4
5	40	5

PROGRAM 1: Histogram of Customer Ages

Create a histogram to represent the distribution of customer ages. Label the axes and title the chart.

```
# Dataset
age <- c(25, 30, 35, 28, 40)

# Histogram
hist(age,
      main = "Distribution of Customer Ages",
      xlab = "Age",
      ylab = "Frequency",
      col = "skyblue")
```



Output 1: Distribution of Customer Ages

Observation: The histogram represents the distribution of the five customer ages in the given dataset.

PROGRAM 2: Pie Chart of Satisfaction Scores

Generate a pie chart to display the overall distribution of customer satisfaction scores and include labels.

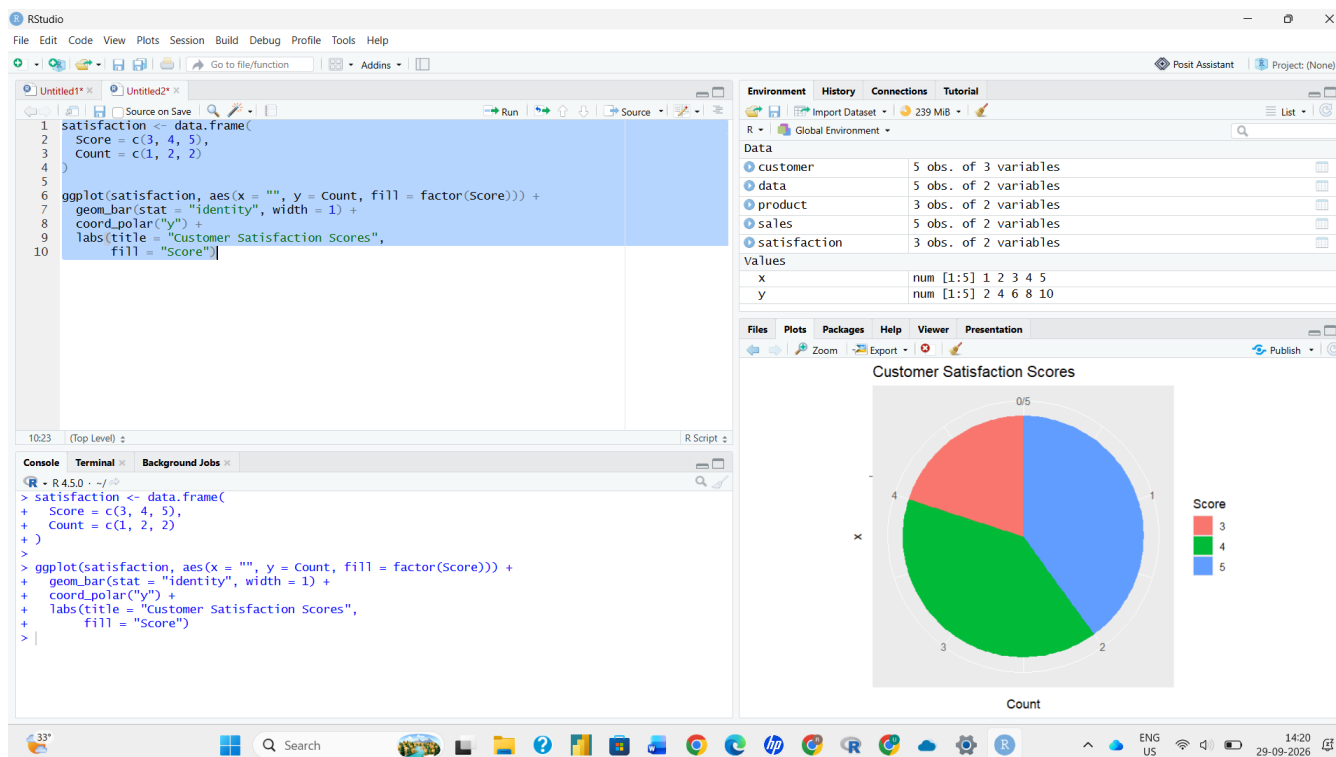
```

# Satisfaction scores
satisfaction <- c(4, 5, 3, 4, 5)

# Calculate frequency
score_count <- table(satisfaction)

# Pie Chart
pie(score_count,
    labels = paste(names(score_count), "Score"),
    main = "Customer Satisfaction Scores")

```



Output 2: Customer Satisfaction Scores Pie Chart

Observation: Scores 4 and 5 each occur twice, while score 3 occurs once.

PROGRAM 3: Stacked Bar Chart by Age Group

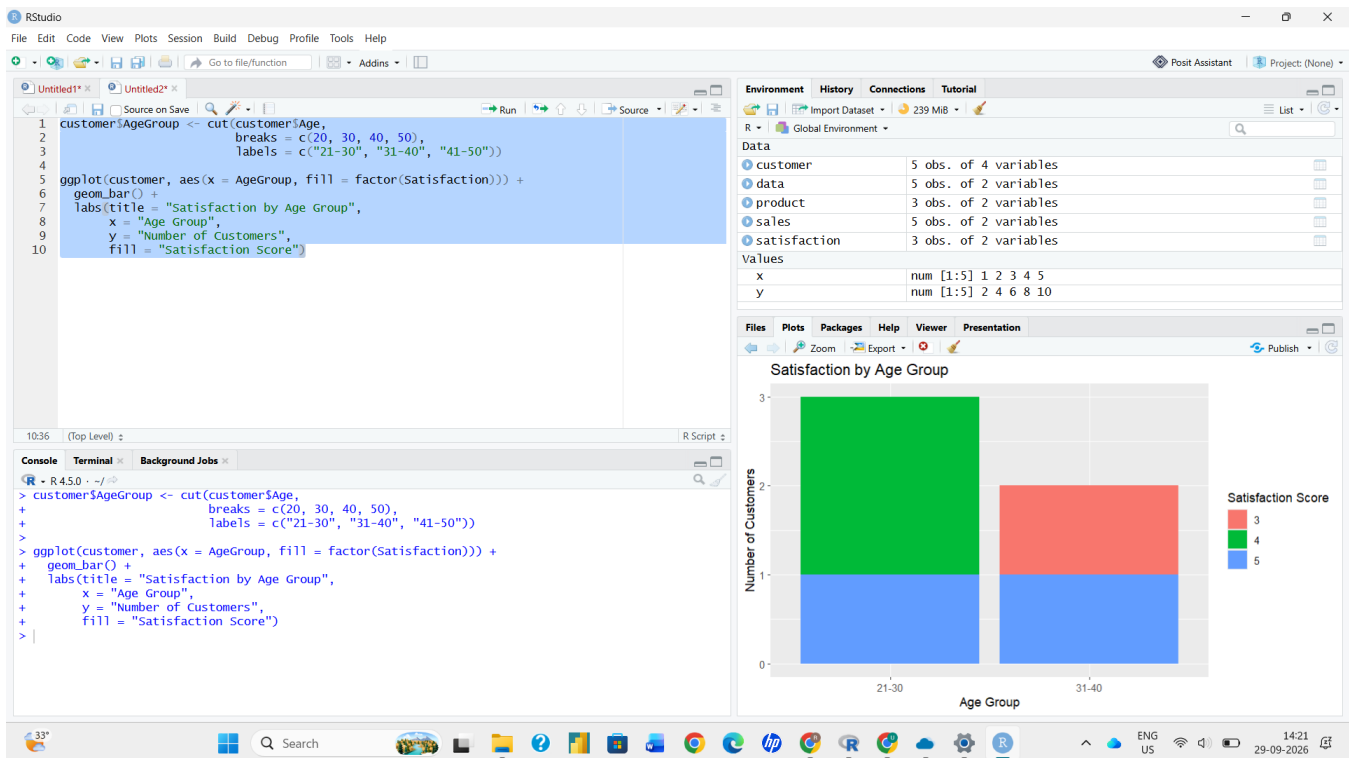
Build a stacked bar chart to visualize the distribution of customer satisfaction scores by age group.

```
# Dataset
age <- c(25, 30, 35, 28, 40)
satisfaction <- c(4, 5, 3, 4, 5)

# Create age groups
age_group <- cut(age,
  breaks = c(20, 29, 39, 49),
  labels = c("20-29", "30-39", "40-49"))

# Create table
data <- table(age_group, satisfaction)

# Stacked Bar Chart
barplot(data,
  beside = FALSE,
  main = "Satisfaction Scores by Age Group",
  xlab = "Age Group",
  ylab = "Number of Customers",
  col = c("skyblue", "lightgreen", "orange"),
  legend.text = TRUE)
```



Output 3: Satisfaction Scores by Age Group

Observation: The code creates three age groups and stacks the satisfaction-score frequencies for each group.

Note: The supplied RStudio screenshot is inserted exactly as captured, including its displayed x-axis labels.

PROGRAM 4: Word Cloud of Customer Feedback

Develop a word cloud from open-ended customer feedback to identify prevalent customer sentiments.

```

library(wordcloud)
library(RColorBrewer)

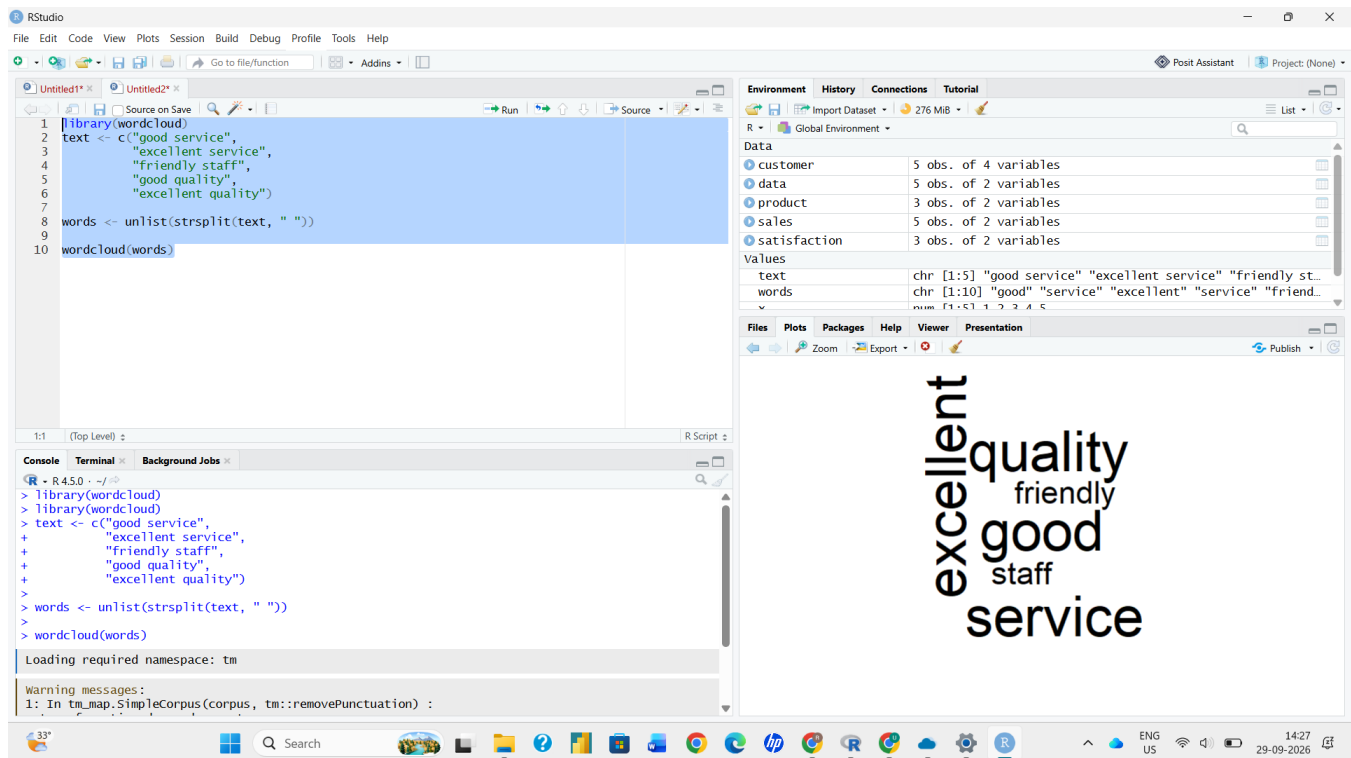
feedback <- c(
  "good service",
  "excellent service",
  "good experience",
  "friendly staff",
  "excellent service"
)

words <- unlist(strsplit(tolower(feedback), " "))

word_count <- table(words)

wordcloud(names(word_count),
  freq = word_count,
  min.freq = 1,
  scale = c(3, 0.5),
  colors = brewer.pal(8, "Dark2"),
  random.order = FALSE)

```



Output 4: Customer Feedback Word Cloud

Observation: The word cloud highlights frequently occurring terms in the supplied sample feedback, including service, excellent, good, friendly, and experience.

Set 2 Procedure

1. Create vectors for customer age and satisfaction score.
2. Use `hist()` to represent the distribution of customer ages.
3. Use `table()` and `pie()` to display the satisfaction score distribution.
4. Create age groups using `cut()`, form a frequency table, and use `barplot()` for the stacked bar chart.
5. Prepare open-ended feedback text, count the words, and use `wordcloud()` to visualize frequent terms.

Set 2 Result

The customer satisfaction data was successfully visualized using a histogram, pie chart, stacked bar chart, and word cloud.